Secure Multi-Region Hub-and-Spoke Microservices Architecture on AWS

The main Problem

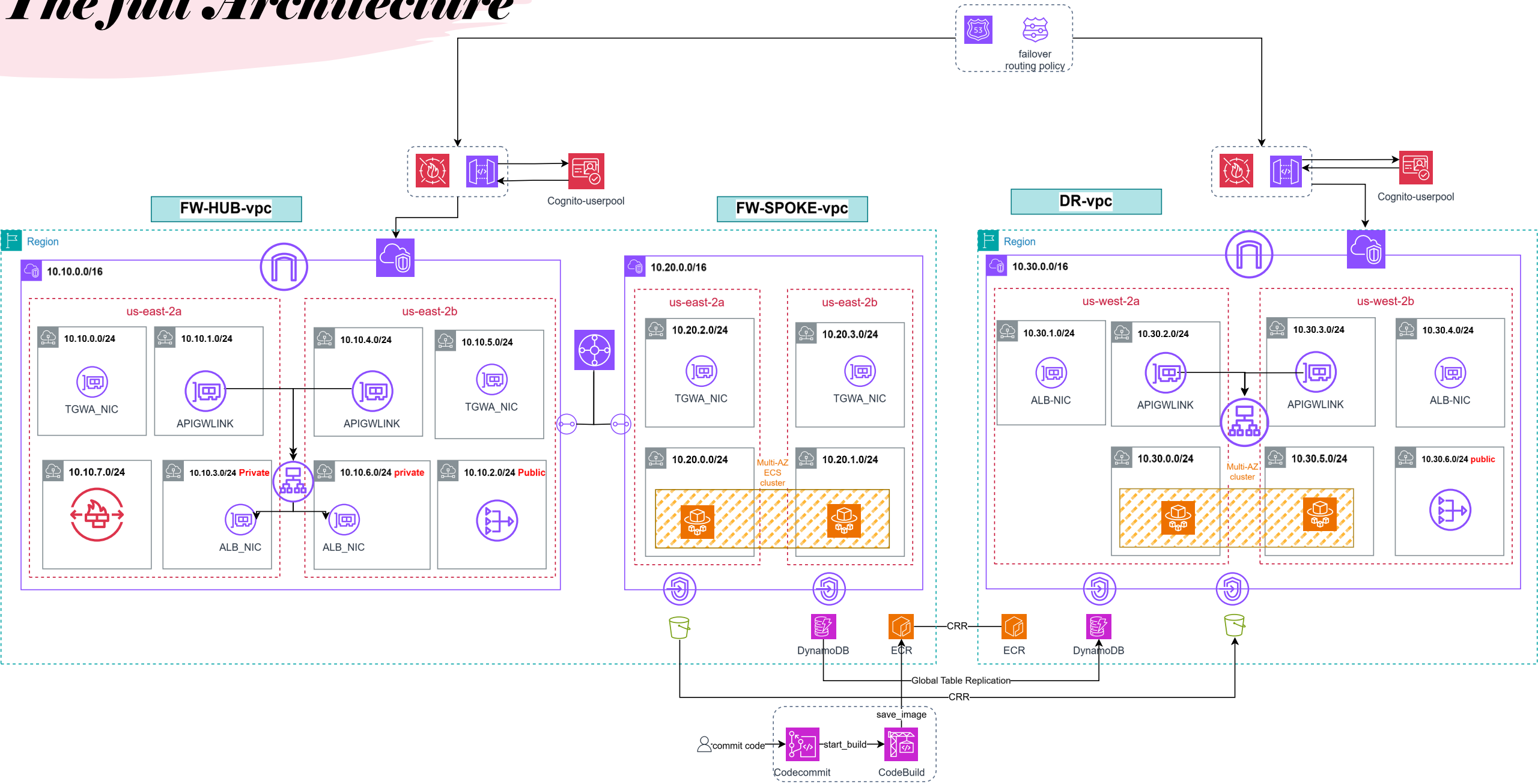
HOW DO WE SECURELY RUN A HIGHLY AVAILABLE APPLICATION

ACROSS MULTIPLE ENVIRONMENTS (VPCS)

WITH CENTRALIZED SECURITY AND DISASTER RECOVERY

WITHOUT EXPOSING INTERNAL SERVICES TO THE INTERNET?

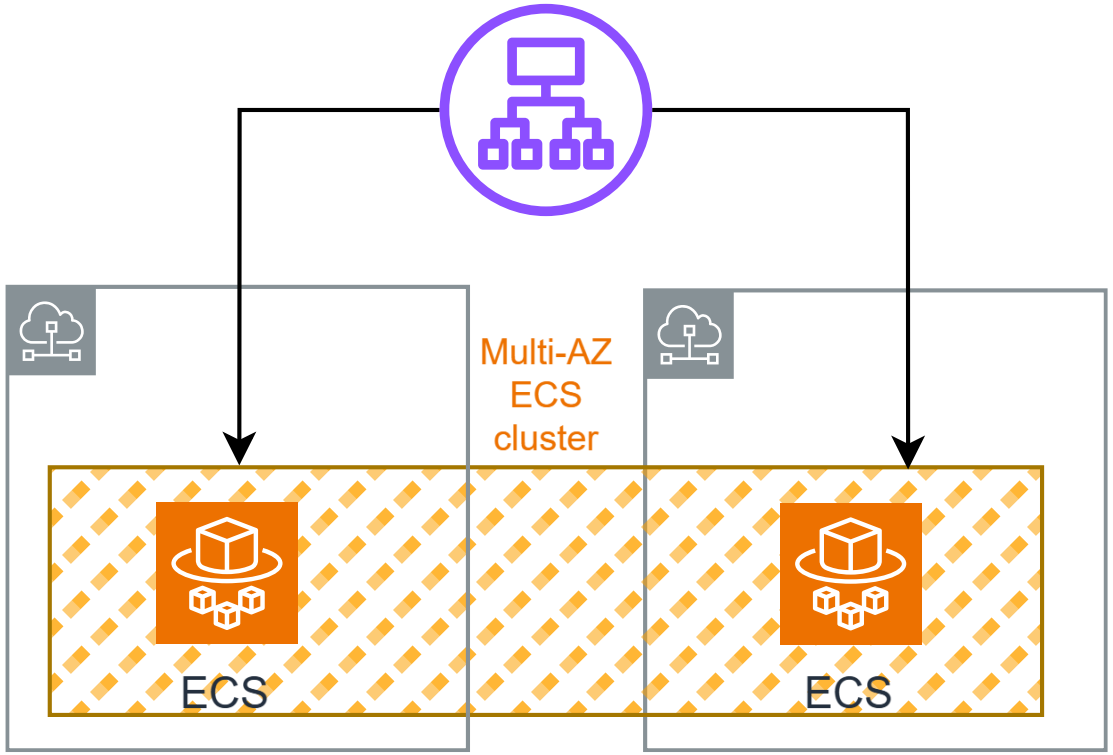
The full Architecture



Key Technical Challenges Pt. 1

- Standardized, Portable, and Independently Scalable Application Deployment
- Applications fail under traffic spikes without automatic scaling
- Security is scattered and hard to control
- Exposing services publicly increases risk
- Entire region failure = total outage

ALB & ECS



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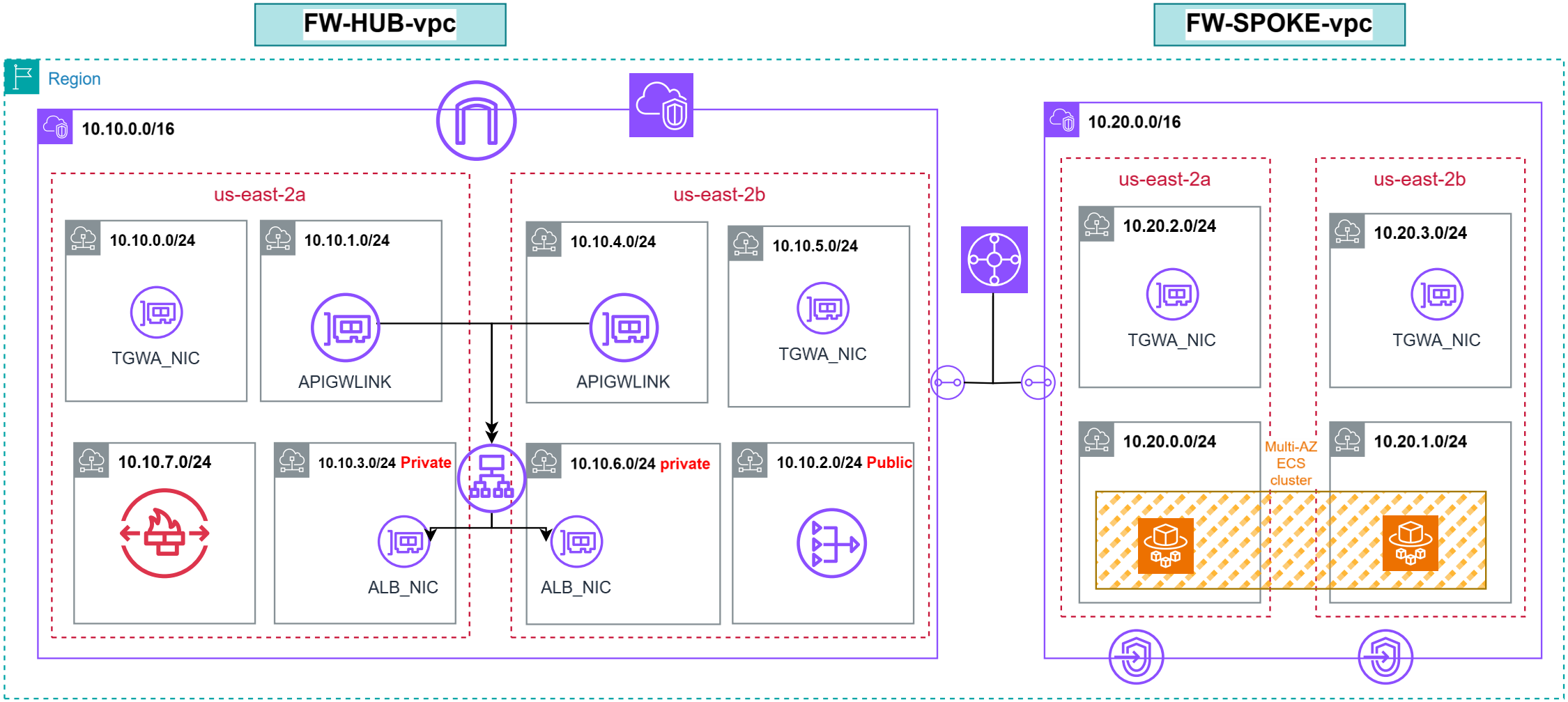
HUB & SPOKE


FW-HUB-vpc
FW-SPOKE-vpc
Region
10.10.0.0/16
us-east-2a
us-east-2b
10.10.0.0/24
TGWA_NIC
10.10.1.0/24
APIGWLINK
10.10.4.0/24
APIGWLINK
10.10.5.0/24
TGWA_NIC
10.10.7.0/24
10.10.3.0/24 Private
ALB_NIC
10.10.6.0/24 private
ALB_NIC
10.10.2.0/24 Public
10.20.0.0/16
us-east-2a
us-east-2b
10.20.2.0/24
TGWA_NIC
10.20.3.0/24
TGWA_NIC
10.20.0.0/24
10.20.1.0/24
Multi-AZ
ECS
cluster

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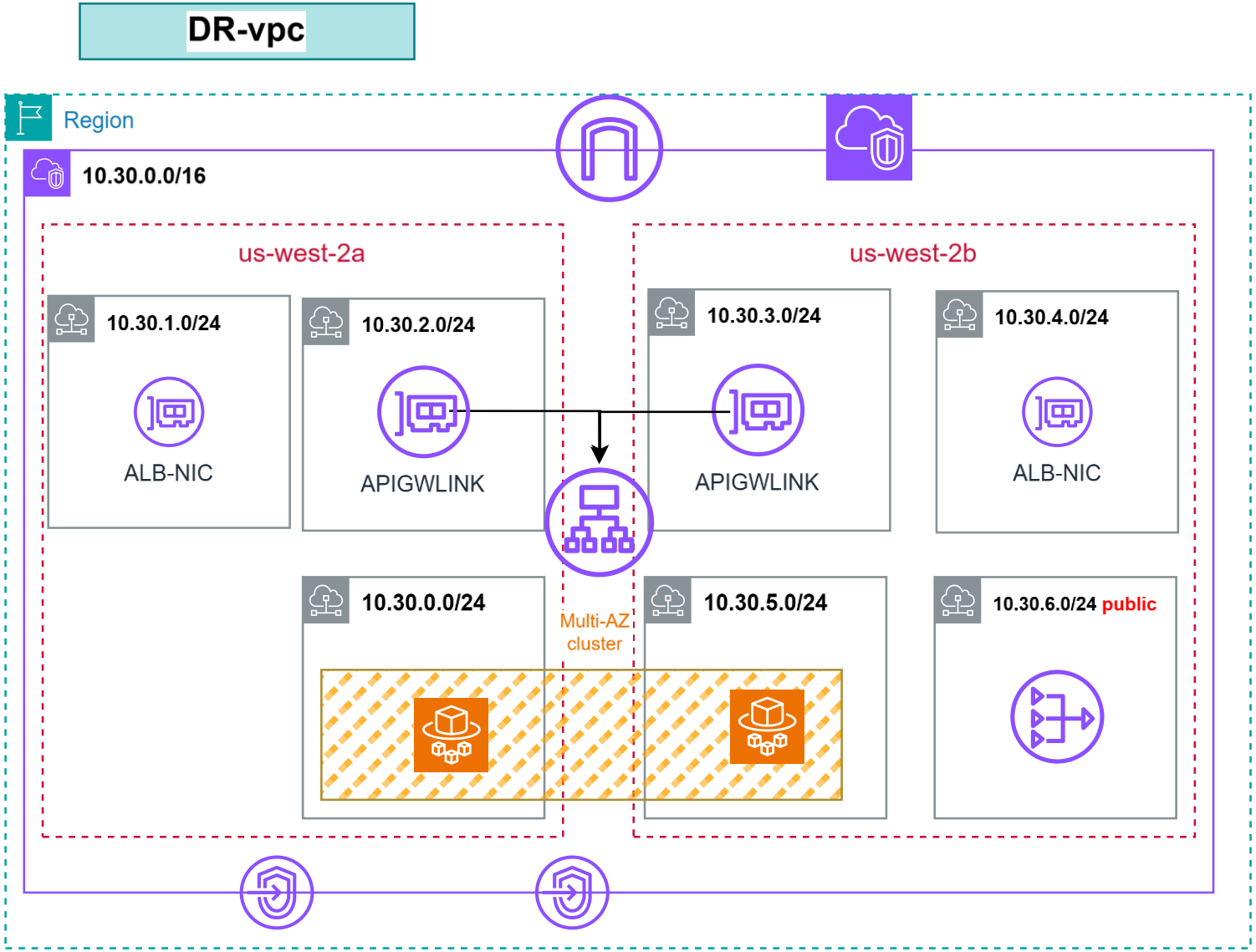
Disaster Recovery



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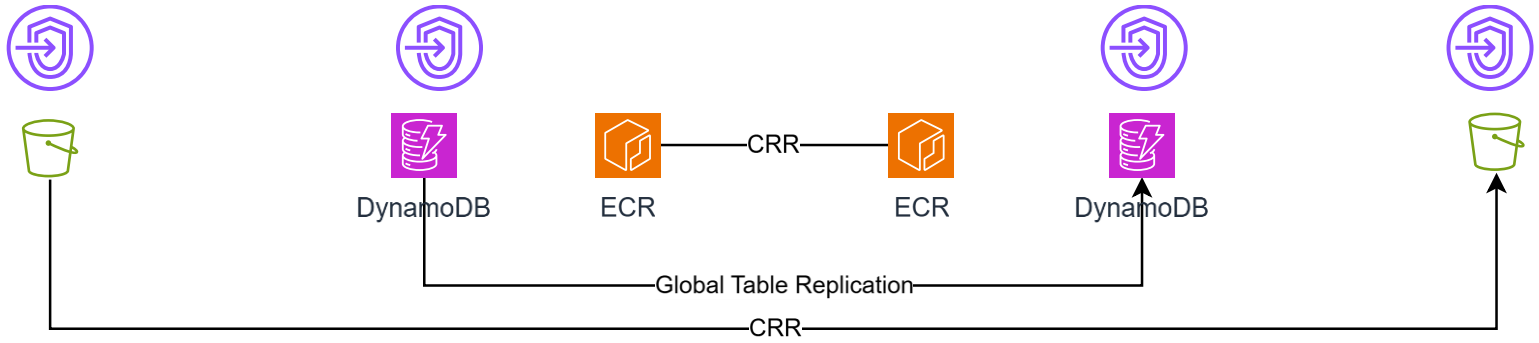
Key Technical Challenges Pt. 2

- Data inconsistency across regions
- Request management & control
- Direct service-to-service communication becomes difficult to manage at scale
- the need to Decouple clients from services
- Achieving Release Governance by decoupling immutable artifact creation from orchestrated deployment through a Manual Approval Gate

Data Replication



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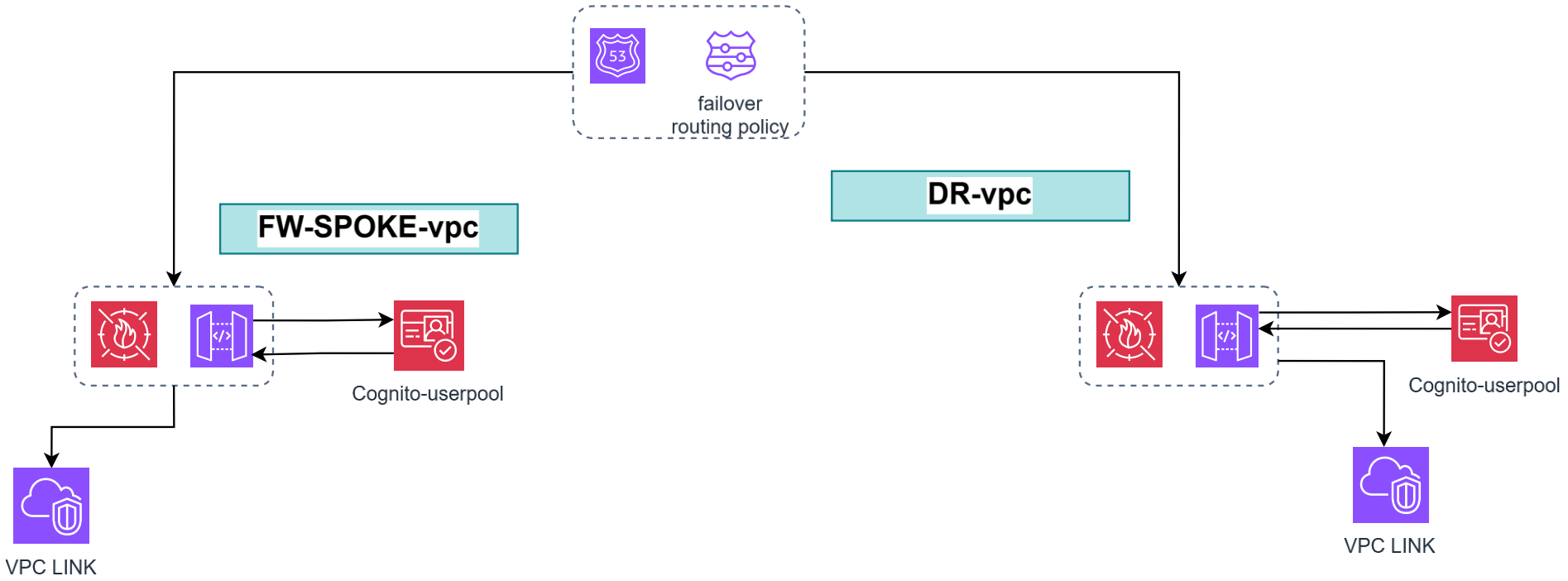
The front door



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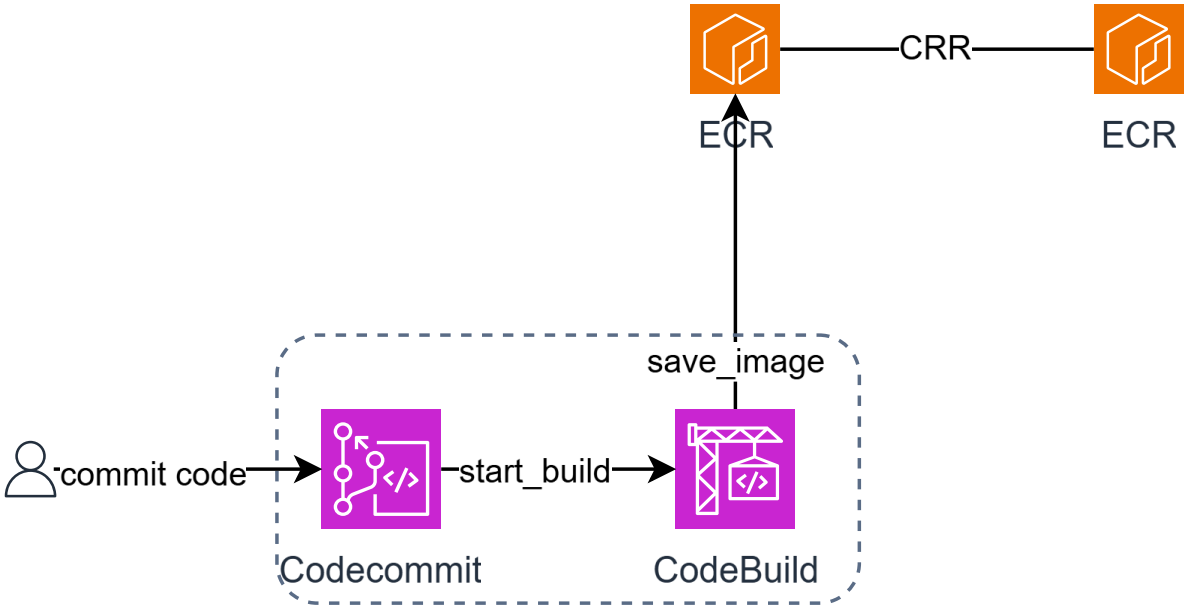
Governed Release



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